

Class 10 Mathematics: Polynomials & Quadratic Equations Guide

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Chapter 1: Overview & Key Definitions

A polynomial is an expression consisting of variables and coefficients, that involves only the operations of addition, subtraction, multiplication, and non-negative integer exponentiation of variables. An example of a polynomial of a single variable x is $x^2 - 4x + 7$.

Chapter 2: Quadratic Equations Formula Matrix

Concept	Mathematical Formula	Key Description
Standard Form	$ax^2 + bx + c = 0$	Quadratic equation standard notation ($a \neq 0$)
Discriminant	$D = b^2 - 4ac$	Determines the nature of roots
Quadratic Formula	$x = \frac{-b \pm \sqrt{D}}{2a}$	Universal algebraic formula for finding roots
Sum of Roots	$\alpha + \beta = -b / a$	Relationship between coefficients and roots
Product of Roots	$\alpha \cdot \beta = c / a$	Relationship between constant term and roots

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